INCH-POUND

MIL-M-28787/47B
AMENDMENT 3
28 April 1993
SUPERSEDING
AMENDMENT 2
19 November 1990

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

BUFFER, THREE-STATE, DIGITAL, KEY CODE BBB

This amendment forms a part of Military Specification MIL-M-28787/47B(EC), dated 7 March 1980, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 6

TABLE II: Delete and substitute new table II as printed on page 4 of this amendment.

PAGE 10

4.3: Delete and substitute the following:

"4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified."

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*4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through GG shall be performed in any order with the following exceptions. Test conditions A, B, C, D, E, F, and G are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D, E, F, or G are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, F, and G where applicable, followed by the remaining tests in the sequence."

*4.5.1, fifth line: Delete "E through EE" and substitute "H through EE."

*4.5.3: Delete and substitute the following:

"4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests H through EE shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in

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table II. Following the return of the module to 25 ± 5°C, electrical tests H through EE shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table II."

6.3: Delete and substitute the following:

"6.3 Cross reference of classification. Military Specification MIL-M-28787/47 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833638. Part number M28787/47-1 supersedes part numbers 2833638 and M28787/47."

PAGE 13

*FIGURE 3: Delete and substitute new figure 3 as printed on page 5 of this amendment.

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FIGURE 6, NOTE 4, second line: Delete "minute" and substitute "second".

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FIGURE 7: Delete and substitute new figure 7 as printed on page 6 of this amendment.

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FIGURE 10: Delete and substitute new figure 10 as printed on page 7 of this amendment.

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TABLE XI, Test condition M2, A to pin column: Delete "40" and substitute "38".

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FIGURE 13: Delete and substitute new figure 13 as printed on page 8 of this amendment.

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TABLE XVI, Test condition T14, A to pin column: Add "20" in column.

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TABLE XIX, Test condition AA5, C to pin column: Add "35" between 34 and 38.

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TABLE XIX, Test condition BB9, C to pin column: Add "35, 38, 40" after 34.

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*FIGURE 32, ITEM C01 box: Delete "C01" and substitute "C06".

*FIGURE 32, NOTE 3, first line: Delete "(C01)" and substitute "(C06)".

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*FIGURE 32, NOTE 3, second line: Delete "Frequency = 120 Hz." and substitute "Frequency = 1 KHz."

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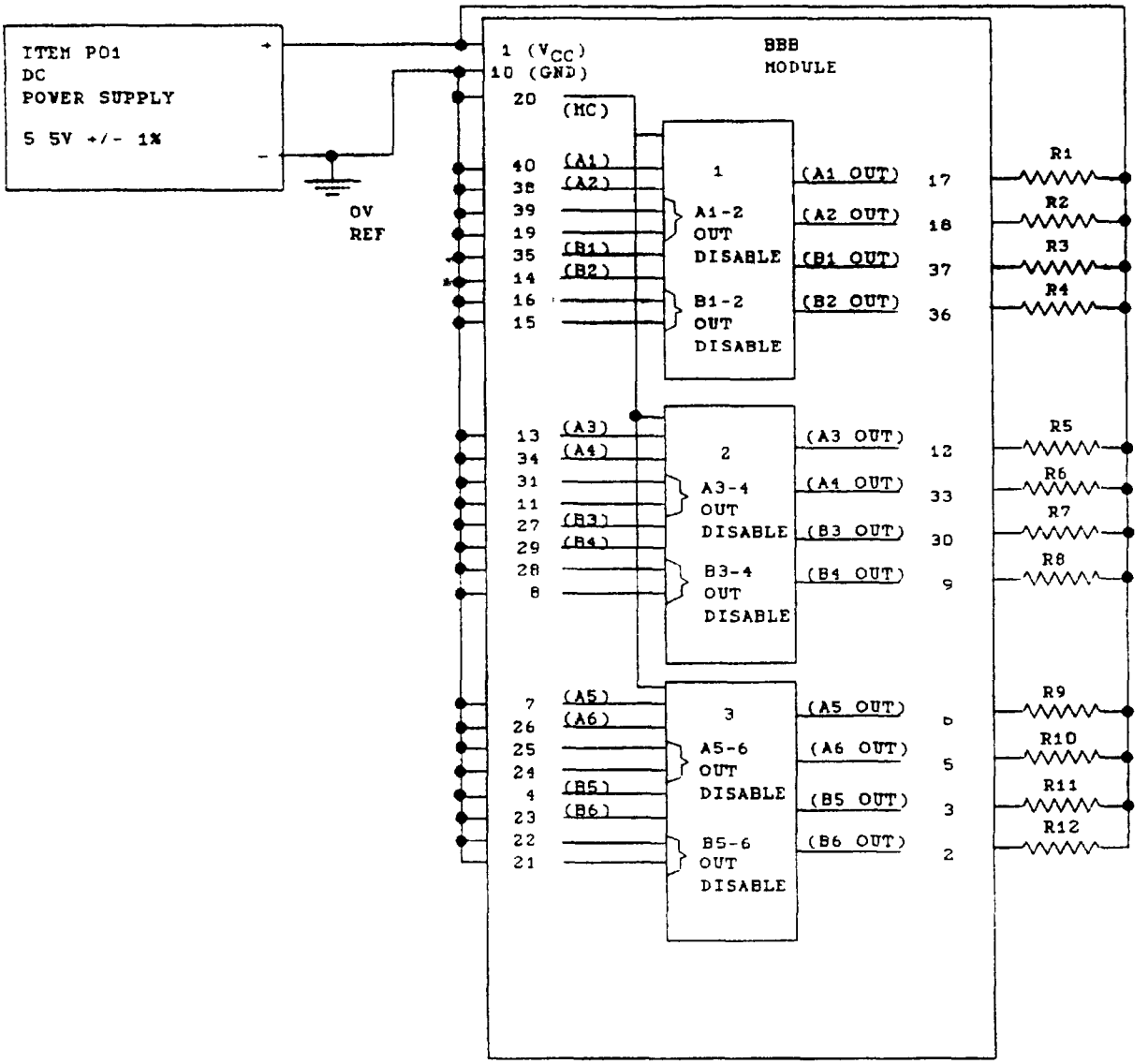
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TABLE II. Module electrical requirements.

Test cond	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25 ± 5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation, frame	III	S	10	-	1	-	10	-	1	-	MΩ
A2	Isolation, unused contacts	III	X	10	-	1	-	10	-	1	-	MΩ
B1-B12	Maximum stress current check	IV	S	-	-	-	-	-	-	-	-	-
C1-C12	Vout (low-loaded)	V	S	0.01	1.0	0.01	1.0	0.01	1.0	0.01	1.0	V
D1-D24	Input breakdown current (data and disable)	VI	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
D25	Input breakdown current (mode control)	VI	S	0.0	3.0	0.0	3.0	0.0	3.0	0.0	3.0	mA
E1-E12	Output short circuit current	VII	S	-40	-120	-40	-120	-40	-120	-40	-120	mA
F1-F25	Input clamp diode voltage	VIII	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
G1-G12	Output clamp diode voltage	VIII	X	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
H1-H12	Output leakage current, (high level)	IX	X	-40	40	-40	40	-40	40	-40	40	µA
J1-J12	Output leakage current, (power off)	IX	X	0	40	0	40	0	40	0	40	µA
K1-K12	Output leakage current, (low level)	X	X	-40	40	-40	40	-40	40	-40	40	µA
L1-L24	Output voltage, low level, Iol = 32 mA	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
L25-L36	Output voltage, low level, Iol = 40 mA	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
M1-M24	Output voltage, high level, Ioh = -2 mA	XI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
M25-M36	Output voltage, high level, Ioh = -40 mA	XI	X	1.8	4.5	1.8	4.5	1.8	4.5	1.8	4.5	V

See footnotes at end of table.

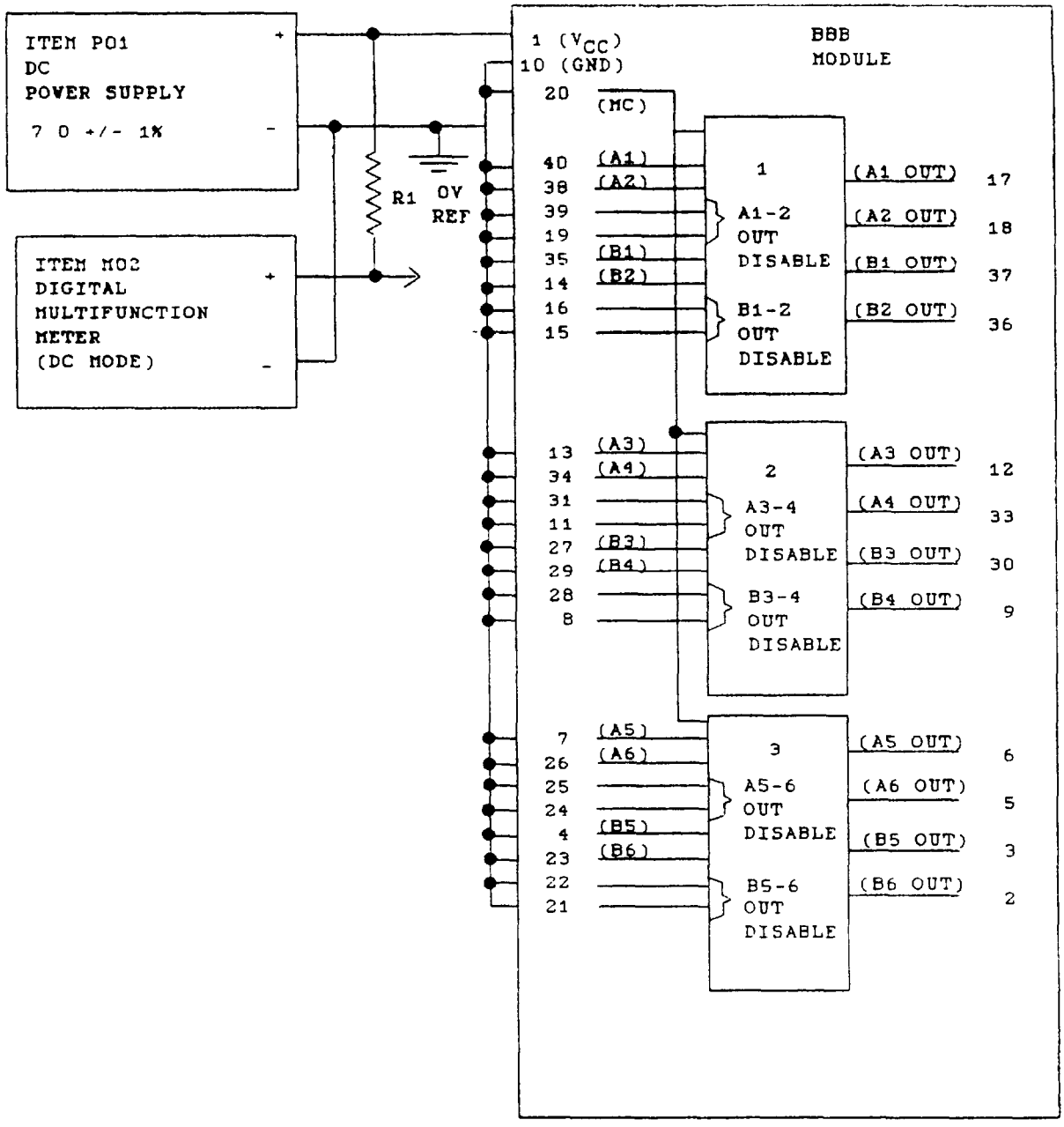
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NOTE: R1-R12 = 127Ω ± 1%, 1/4 W.

FIGURE 3. Transient thermal test circuit.

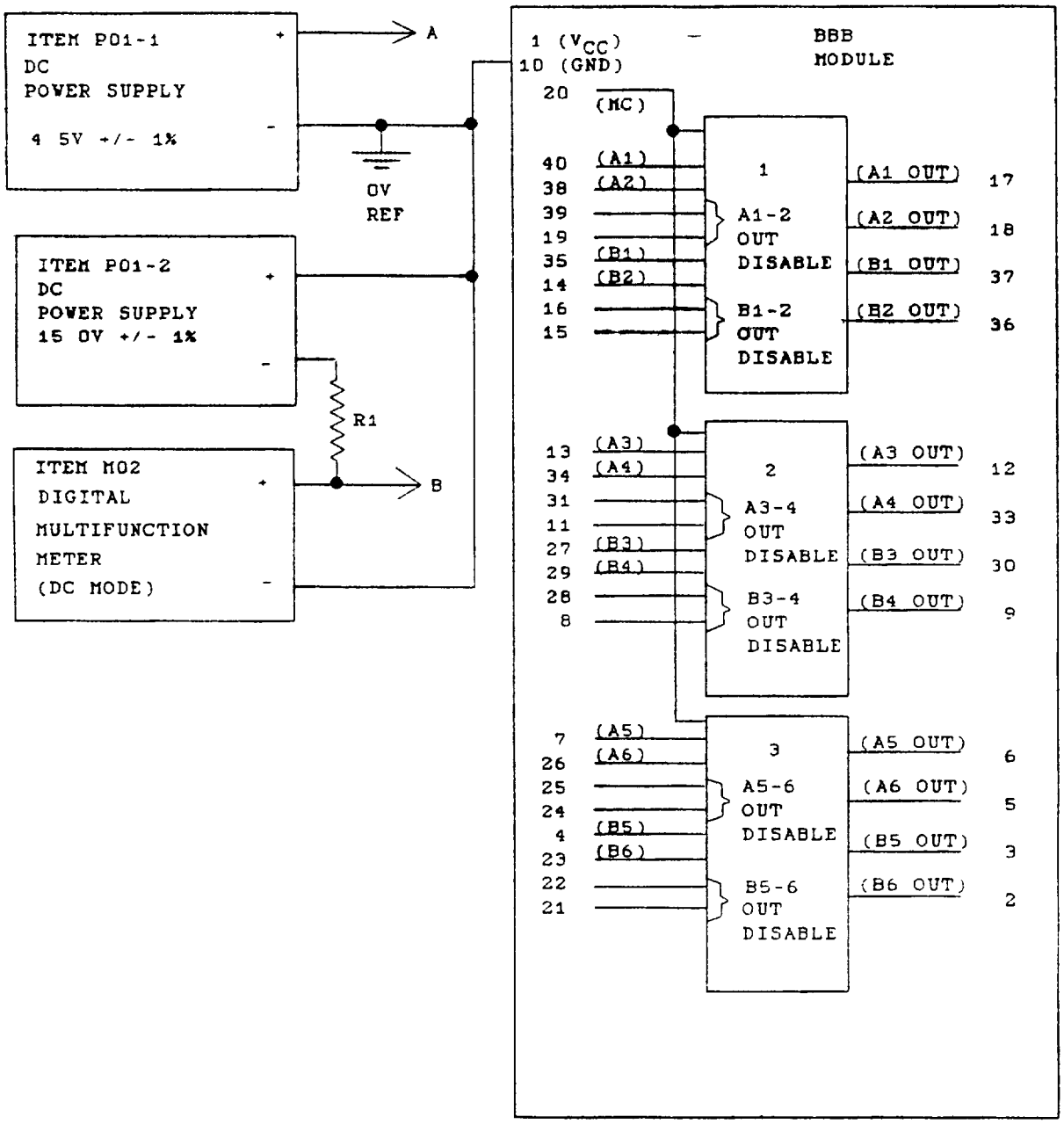
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NOTE: R1 = 127Ω ± 1%, 1/2 W.

FIGURE 7. V_{OUT} (low-loaded) test circuit for table V.

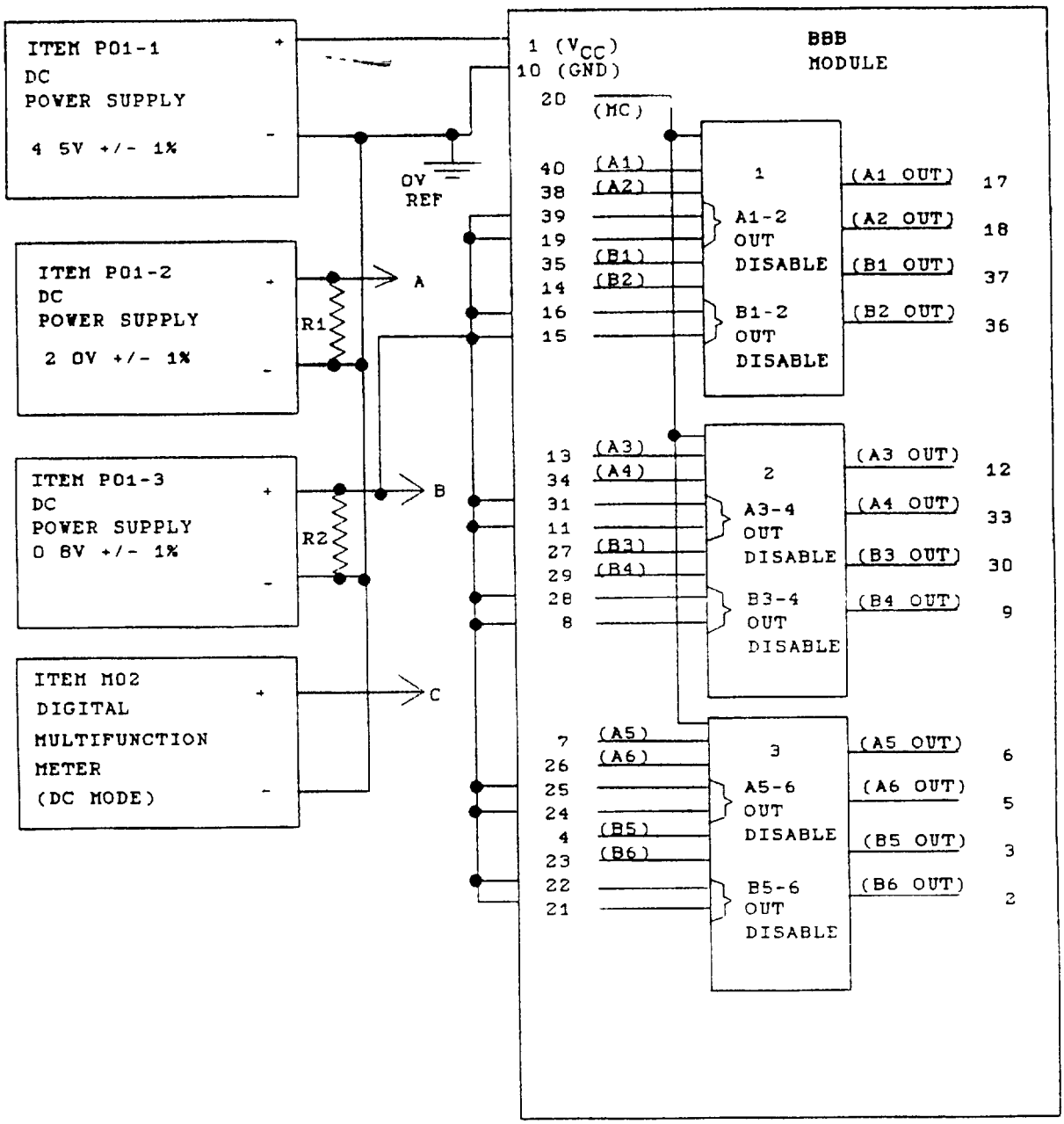
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NOTE: R1 = 1.13kΩ ± 1%, 1/4 W.

FIGURE 10. Input and output clamp diode voltage test circuit for table VIII.

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NOTES:
1. R1 = 51Ω ± 5%, 1/2 W.
2. R2 = 10Ω ± 5%, 1/2 W.
3. Load output under test with load as specified on figure 14.

FIGURE 13. Output voltage (low-level and high-level) test circuit for table XI.

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